



TARGET-90

JKBOSE CLASS-11TH

CONIC SECTIONS

One Shot

MATHS

Mehraj sir

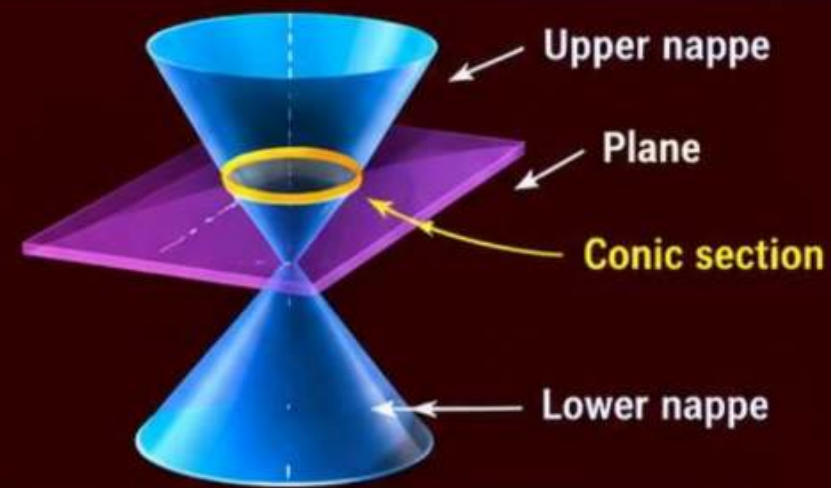


Topics To Be Covered

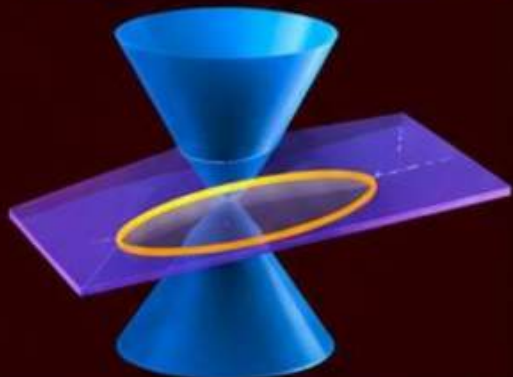
- 1 ... Circle
- 2 ... Parabola
- 3 ... Ellipse + Hyperbola.

CONIC SECTIONS

Conic sections are the curves formed by the intersection of a plane and a right circular cone.



ELLIPSE

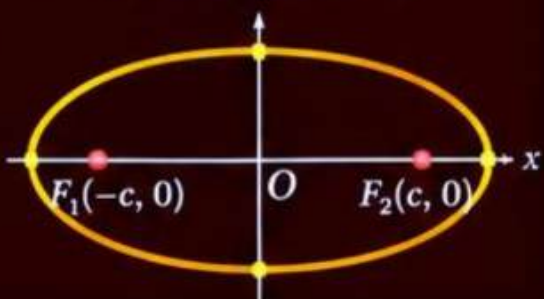


$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$(a > b > 0)$

- Two foci: $(\pm c, 0)$
- Major axis = $2a$
- Minor axis = $2b$
- Eccentricity: $0 < e < 1$
- Sum of distances from the foci is constant.

Shape: Oval



PARABOLA

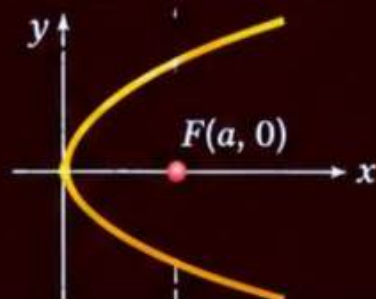


$$y^2 = 4ax$$

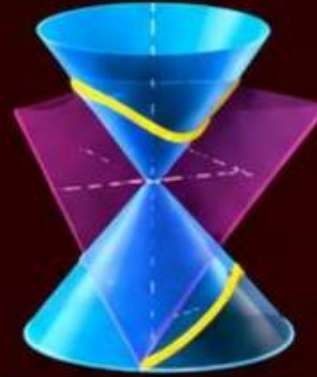
(or $x^2 = 4ay$)

- One focus: $(a, 0)$
- Directrix: $x = -a$
- Eccentricity: $e = 1$
- Distance from focus = distance from directrix.

Shape: U-shaped



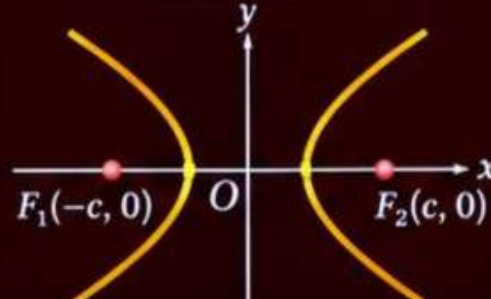
HYPERBOLA



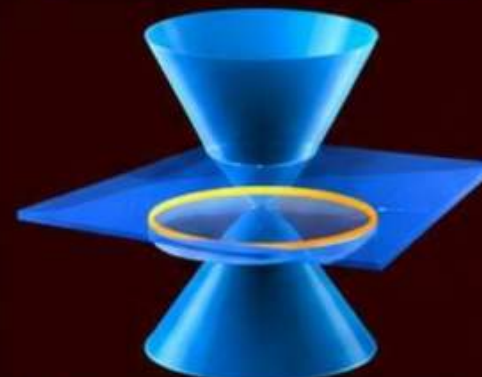
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

- Two foci: $(\pm c, 0)$
- Transverse axis = $2a$
- Conjugate axis = $2b$
- Eccentricity: $e > 1$
- Difference of distances from the foci is constant.

Shape: Open curve



CIRCLE

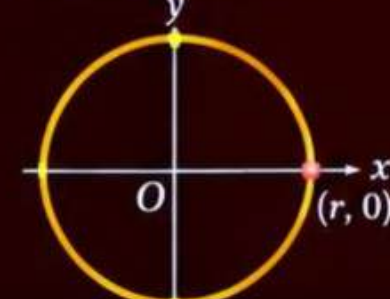


$$x^2 + y^2 = r^2$$

$(a = b = r)$

- One centre: $(0, 0)$
- Radius = r
- Eccentricity: $e = 0$
- Special case of ellipse (when $a = b$).

Shape: Round

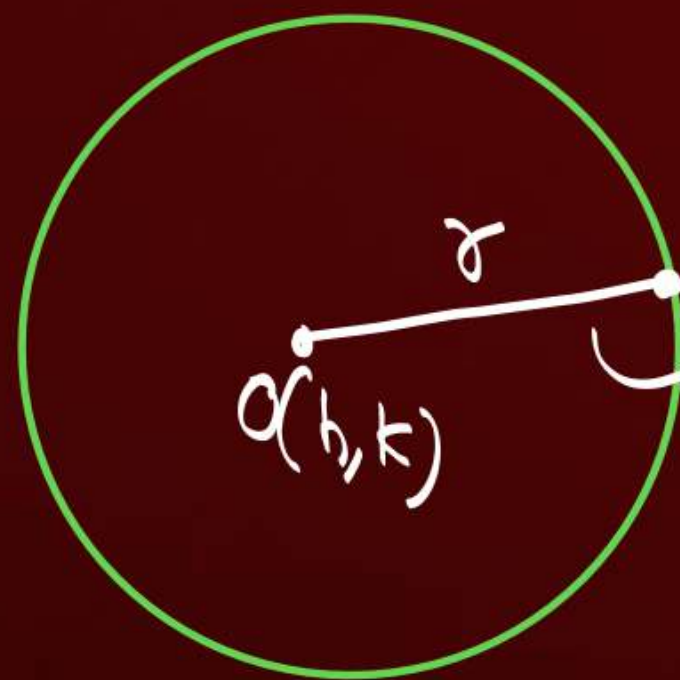
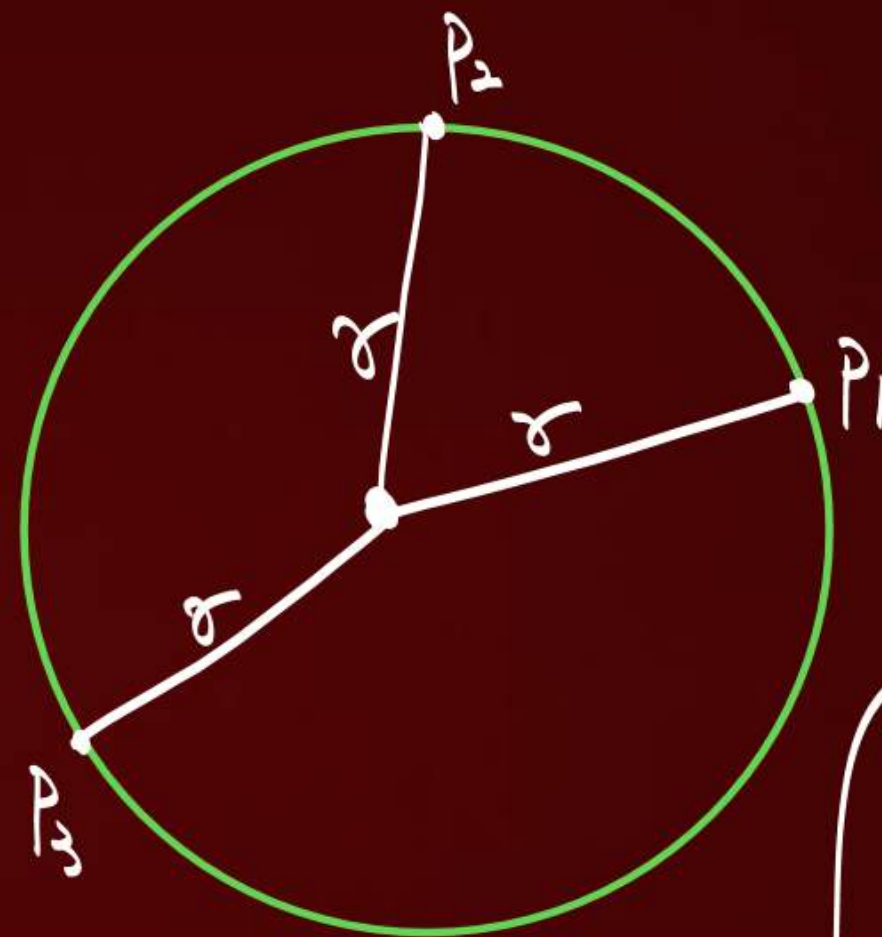


★ Circle



The Collection of points in plane which are equidistant from a fixed point is known as a circle.

* fixed point is known as the Centre of the circle
And the constant distance is known as the radius of the circle



$$\rightarrow (x-h)^2 + (y-k)^2 = r^2$$

$$* x^2 + y^2 + 2gx + 2fy + c = 0 \quad \text{--- (1)}$$

→ Equation of Circle in the general form.

$$\text{Centre} = (-g, -f)$$

$$\text{Radius} = \sqrt{g^2 + f^2 - c}$$

EXAMPLE



Find the equation of the circle with centre $(-3, 2)$ and radius 4.

(Sol) Required circle is:

$$(x+3)^2 + (y-2)^2 = 4^2.$$

$$\Rightarrow x^2 + 9 + 6x + y^2 + 4 - 4y = 16.$$

$$\Rightarrow x^2 + y^2 + 6x - 4y - 3 = 0.$$

QUESTION (NCERT EXAMPLER):

The area of the circle centred at $(1, 2)$ and passing through $(4, 6)$ is.

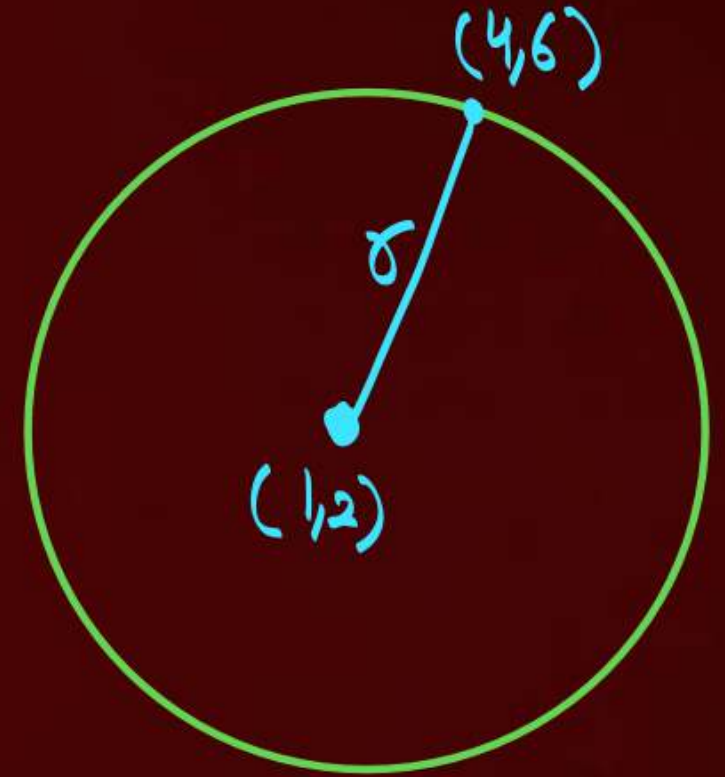
- 1 5π ✗
- 2 10π ✗
- 3 25π ✓
- 4 None of these

$$\text{Here } r = \sqrt{(4-1)^2 + (6-2)^2}$$

$$\Rightarrow r = \sqrt{3^2 + 4^2} = \sqrt{9+16}$$

$$\Rightarrow r = 5$$

$$\text{Area} = \pi(5)^2 = 25\pi \text{ Sq. units}$$



EXAMPLE



Find the centre and the radius of the circle $x^2 + y^2 + 8x + 10y - 8 = 0$.

$$(Sol) \text{ Centre} = (-4, -5)$$

$$\text{Radius} = \sqrt{4^2 + 5^2 - (-8)}$$

$$\Rightarrow \text{Radius} = \sqrt{16 + 25 + 8} = \sqrt{49}$$

$$\Rightarrow \text{Radius} = 7 \text{ units.}$$

QUESTION

In each of the following Exercises 6 to 9, find the centre and radius of the circles.

$$x^2 + y^2 - 8x - 10y - 12 = 0$$

H/w

QUESTION



Find the equation of the circle passing through the points $(4, 1)$ and $(6, 5)$ and whose centre is on the line $4x + y = 16$.

(Sol) Suppose that the required equation of a circle be;

$$(x-h)^2 + (y-k)^2 = r^2 \quad \text{--- (1)}$$

$$\text{Here } \sqrt{(h-4)^2 + (k-1)^2} = \sqrt{(h-6)^2 + (k-5)^2} = r$$

$$\Rightarrow h^2 + 16 - 8h + k^2 + 1 - 2k = h^2 + 36 - 12h + 25 + k^2 - 10k$$

$$\Rightarrow 17 - 8h - 2k = 61 - 12h - 10k$$

$$\Rightarrow 4h + 8k = 44$$

$$\Rightarrow 4(h + 2k) = 44 \Rightarrow h + 2k = 11 \quad \text{--- (2)}$$

Since $O(h, k)$ lies on $4x + y = 16$,

$$\text{then } 4h + k = 16$$

$$\Rightarrow k = 16 - 4h \quad \text{--- (3)}$$

Using (3) in (2), we get;

$$h + 2(16 - 4h) = 11$$

$$\Rightarrow h + 32 - 8h = 11$$

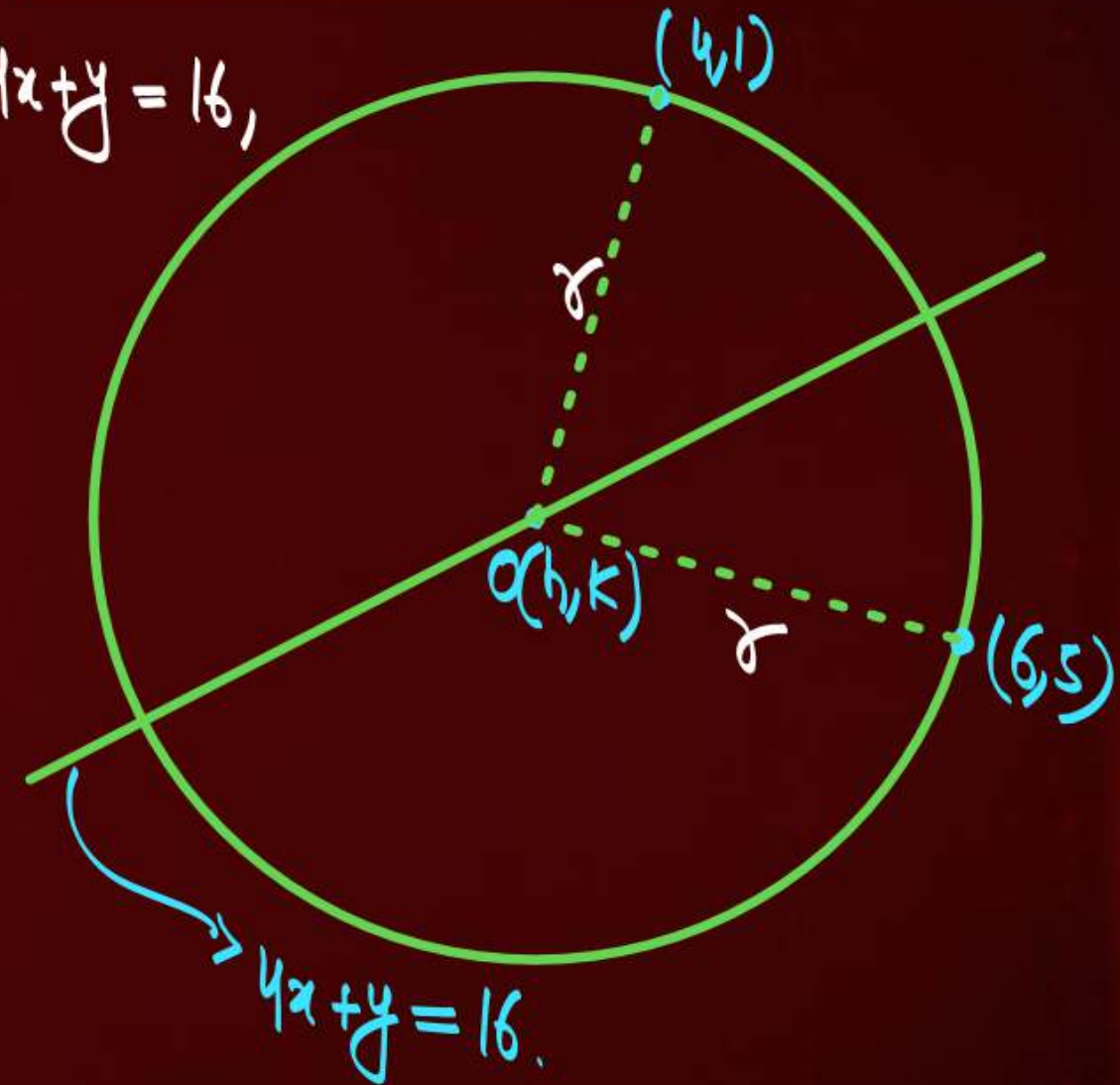
$$\Rightarrow 32 - 7h = 11$$

$$\Rightarrow 7h = 21$$

$$\Rightarrow h = 3$$

$$\text{So } k = 16 - 4 \times 3$$

$$\Rightarrow k = 4$$



$$\Rightarrow \text{Centre} = (h, k) = (3, 4)$$

$$\text{Also } r = \sqrt{(3-6)^2 + (4-5)^2}$$

$$\Rightarrow r = \sqrt{9+1} = \sqrt{10}$$

So from (A), we get

$$(x-3)^2 + (y-4)^2 = \cancel{10}^2$$

$$\Rightarrow x^2 + 9 - 6x + y^2 + 16 - 8y = 10$$

$$\Rightarrow \boxed{x^2 + y^2 - 6x - 8y + 15 = 0}$$

H.P.R.

QUESTION

Find the equation of the circle passing through the points $(2, 3)$ and $(-1, 1)$ and whose centre is on the line $x - 3y - 11 = 0$.

H/W

QUESTION

Find the equation of the circle with radius 5 whose centre lies on x -axis and passes through the point $(2, 3)$.

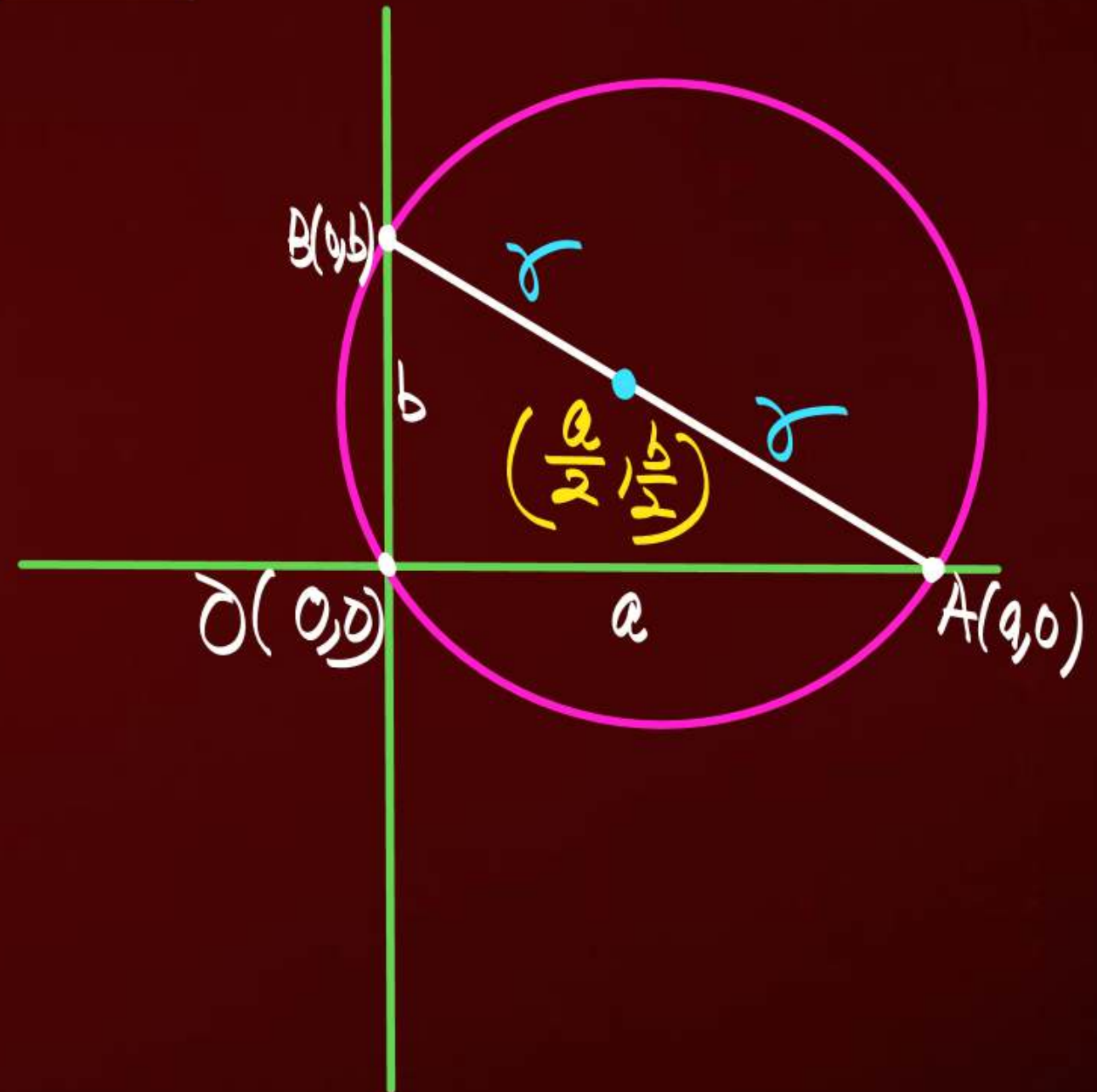
h/w

QUESTION



Find the equation of the circle passing through $(0, 0)$ and making intercepts a and b on the coordinate axes.

$$\left(x - \frac{a}{2}\right)^2 + \left(y - \frac{b}{2}\right)^2 = \frac{a^2 + b^2}{4} \quad (2r)^2 = a^2 + b^2$$
$$\Rightarrow r^2 = \frac{a^2 + b^2}{4}$$

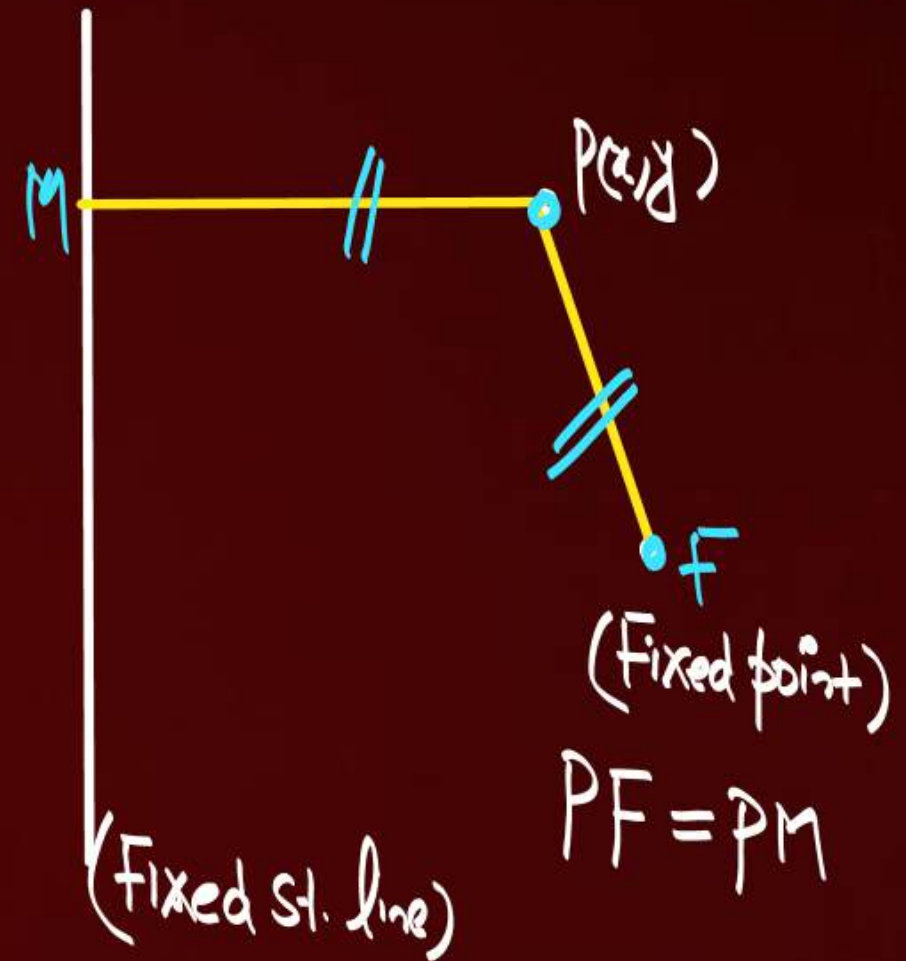
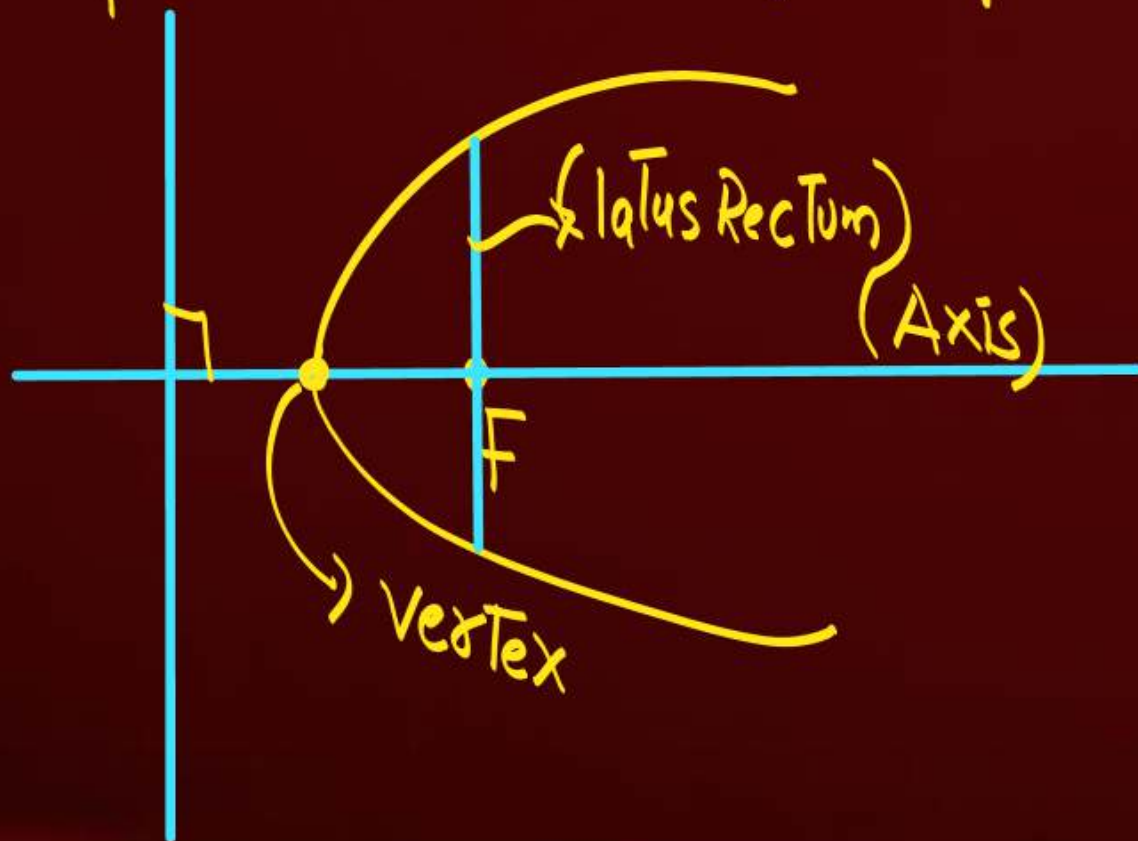


Parabola

The Collection of points in a Plane whose distance from a fixed Point is equal to its distance from a fixed straight line is known as a parabola.

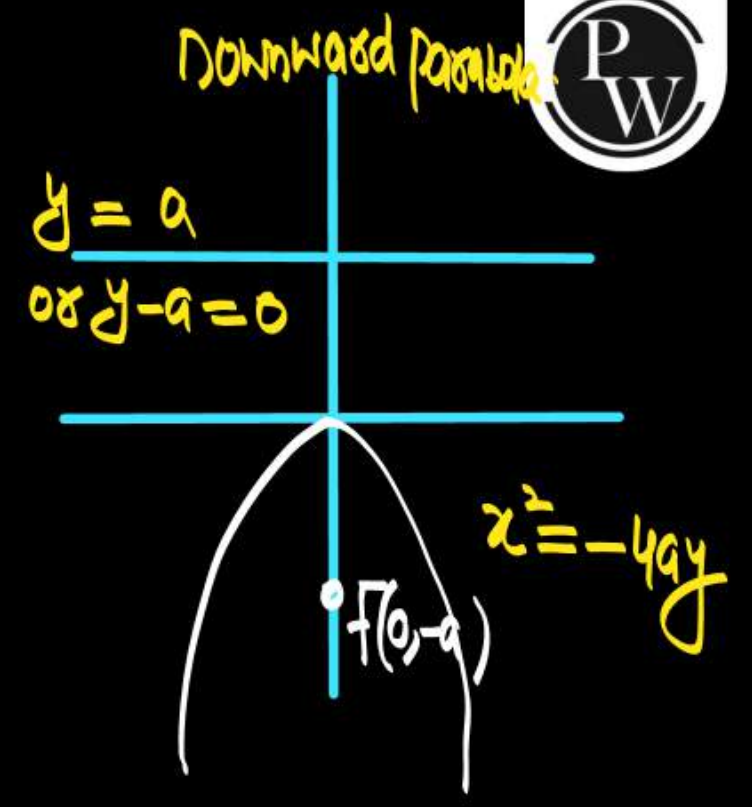
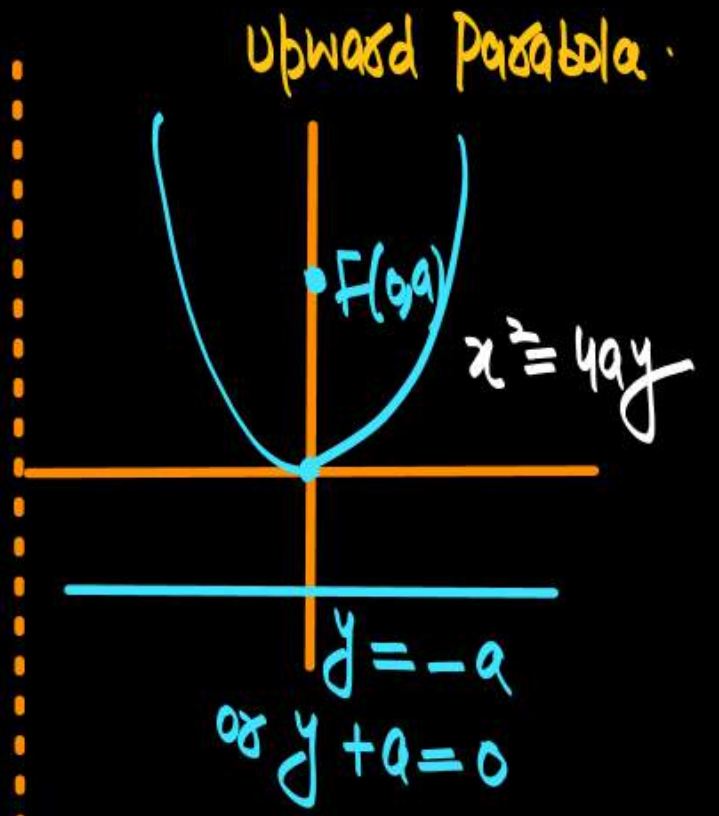
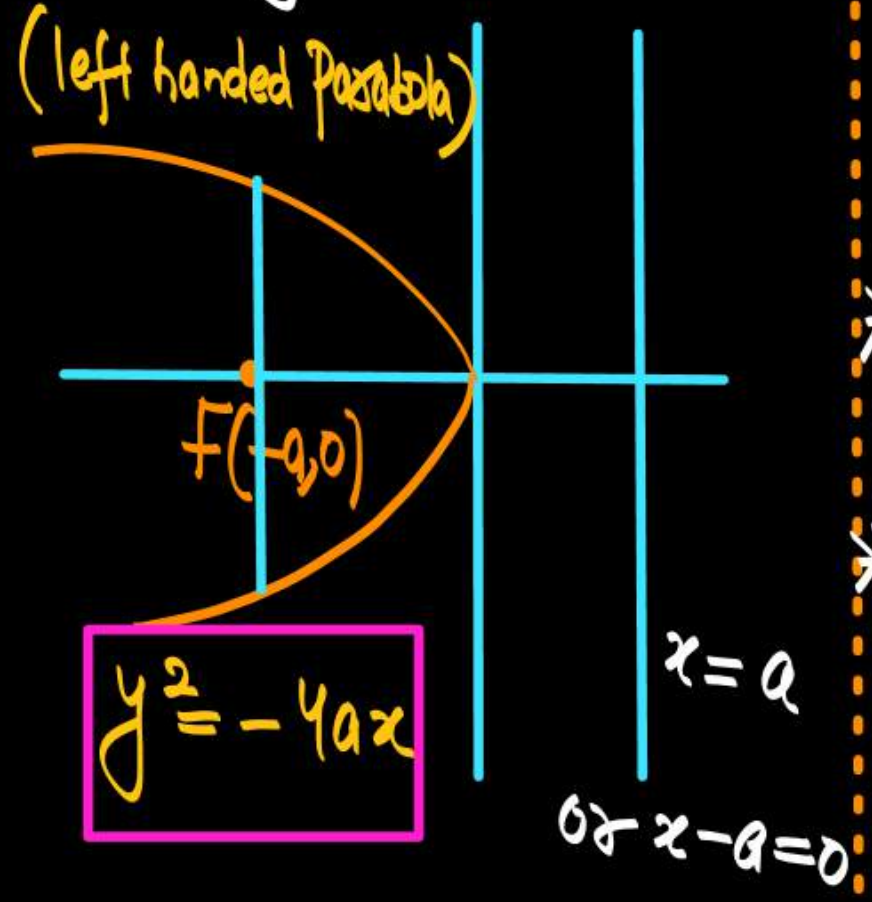
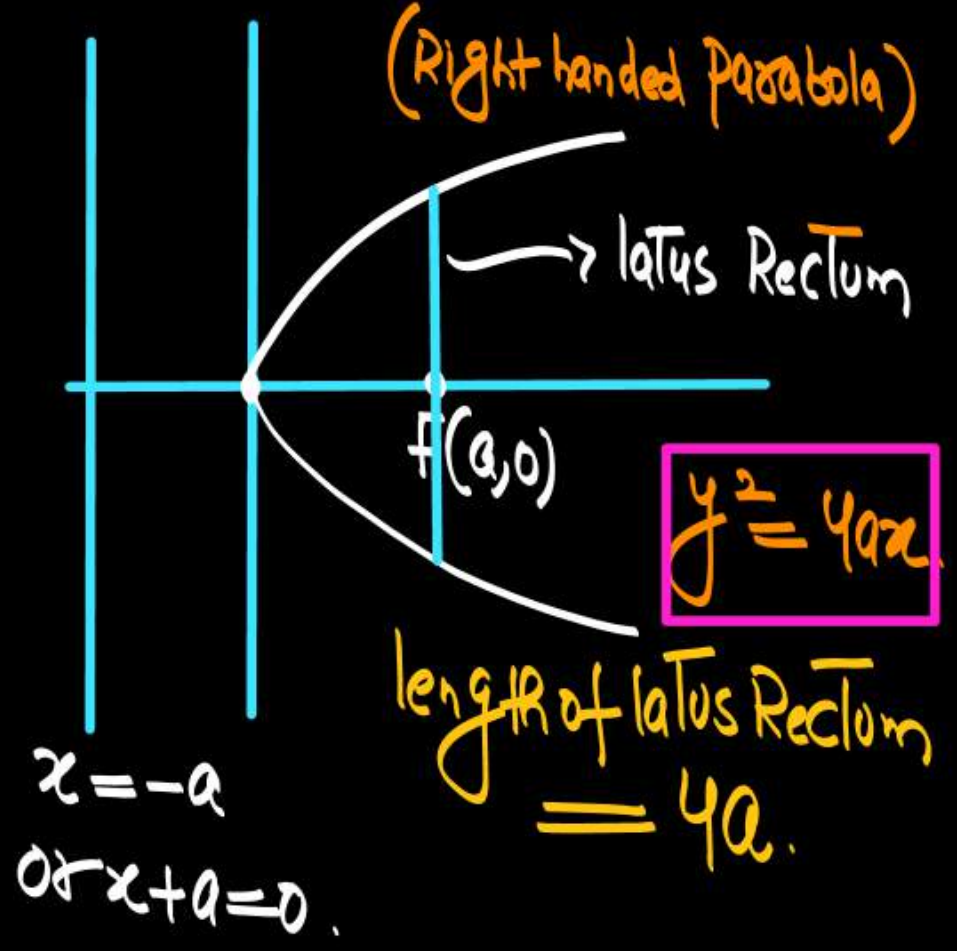
* The fixed straight line is known as the Directrix of the Parabola.

* The fixed point is known as the focus of the Parabola.

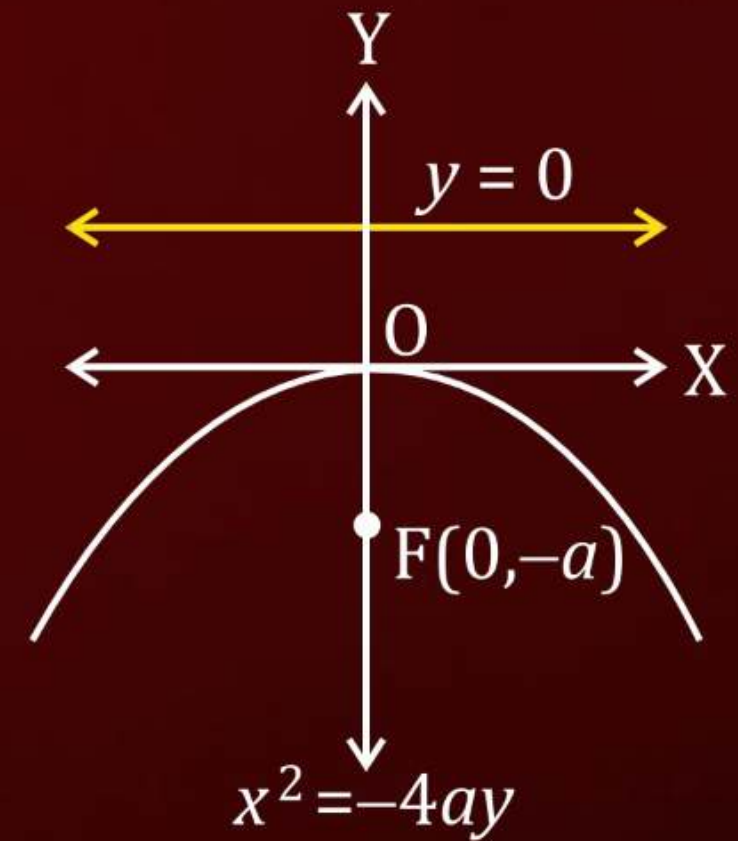
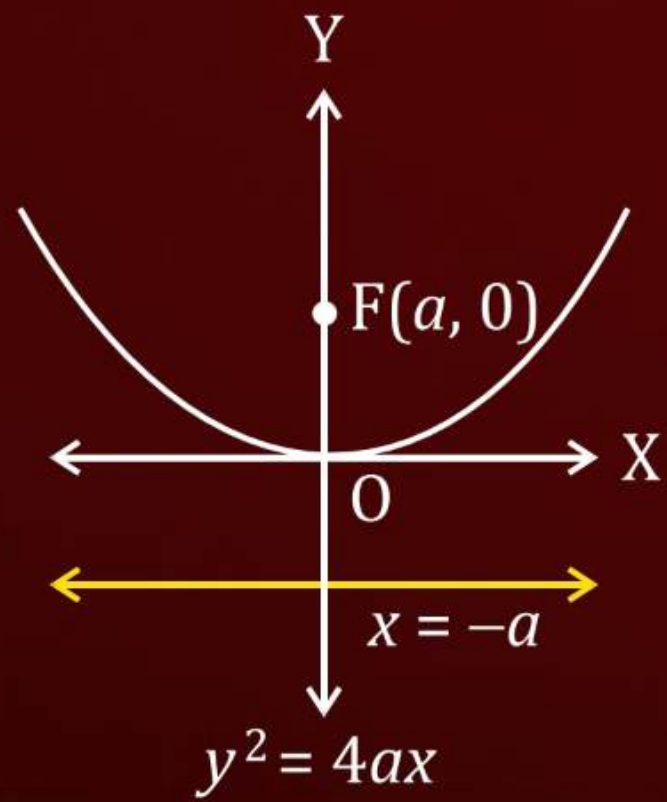
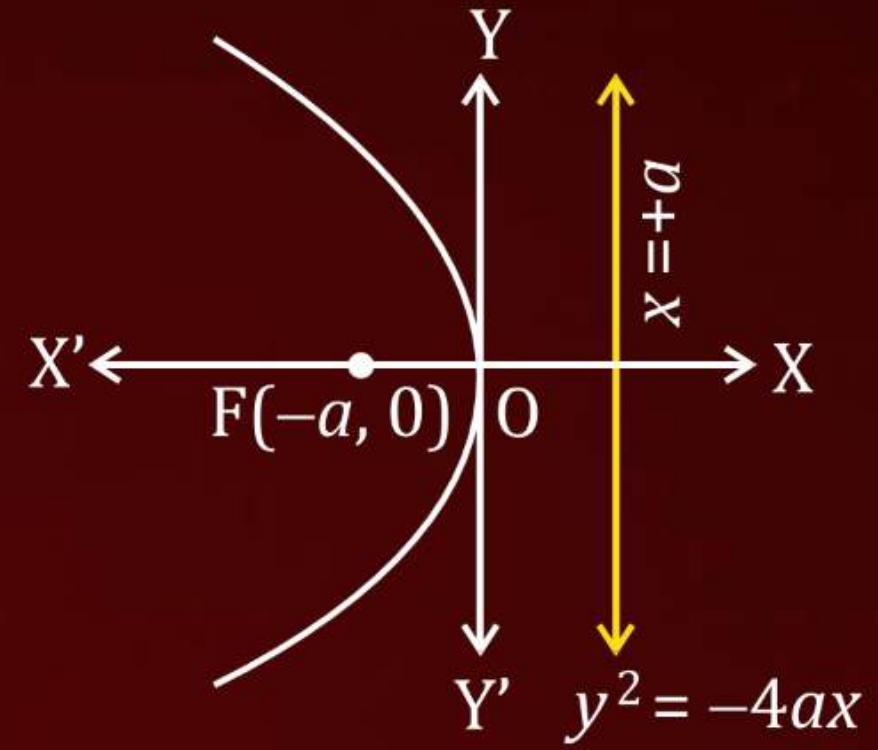
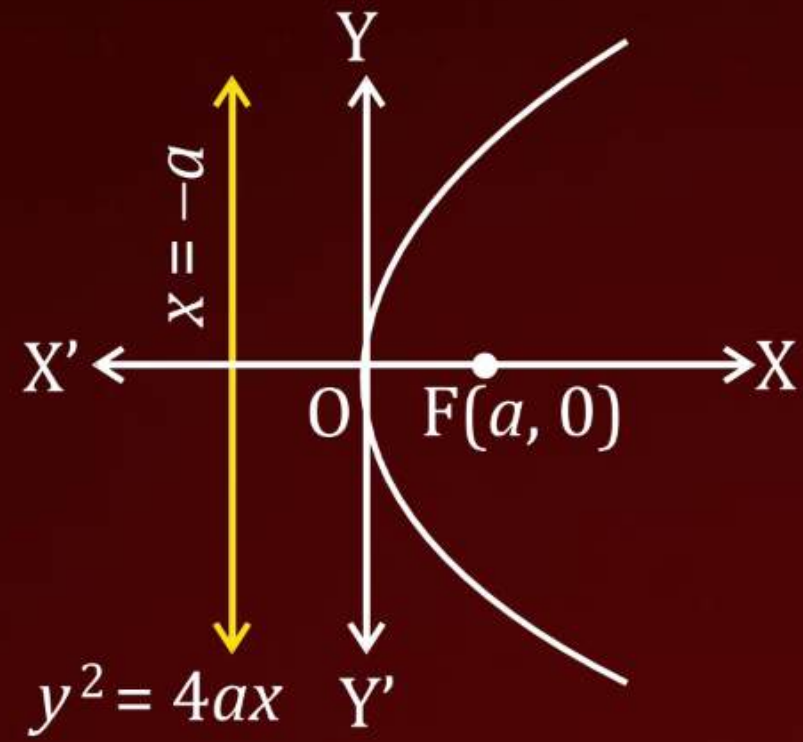


★ Equation of parabola in standard form:

- * Axis of the parabola is along the x -axis or y -axis.
- * Vertex of the parabola is taken at the origin.



- * Upward and downward parabola are symmetric w.r.t. y -axis.
- * Right handed & left handed parabola are symmetric w.r.t. x -axis.



EXAMPLE



Find the coordinates of the focus, axis, the equation of the directrix and latus rectum of the parabola $y^2 = 8x$.

(Sol) Given Parabola is;

$y^2 = 8x$, which is clearly a Right handed Parabola.

Here $4a = 8 \Rightarrow a = 2$.

So, Focus = $(2, 0)$

Axis :- x-axis.

Directrix : $x + 2 = 0$.

length of latus Rectum = $4 \times 2 = 8$.

H.P.R.

EXAMPLE



Find the equation of the parabola which is symmetric about the y -axis, and passes through the point $(2, -3)$.

(Sol) Suppose that the required parabola is:

$$x^2 = -4ay \quad \text{--- (1)}$$

Since $(2, -3)$ lies on (1), so we get:

$$(2)^2 = -4a(-3).$$

$$\Rightarrow 4 = 12a.$$

$$\Rightarrow a = \frac{1}{3}.$$

So, the required equation is:

$$x^2 = -\frac{4}{3}y$$

$$\Rightarrow 3x^2 + 4y = 0$$

Hence the result.

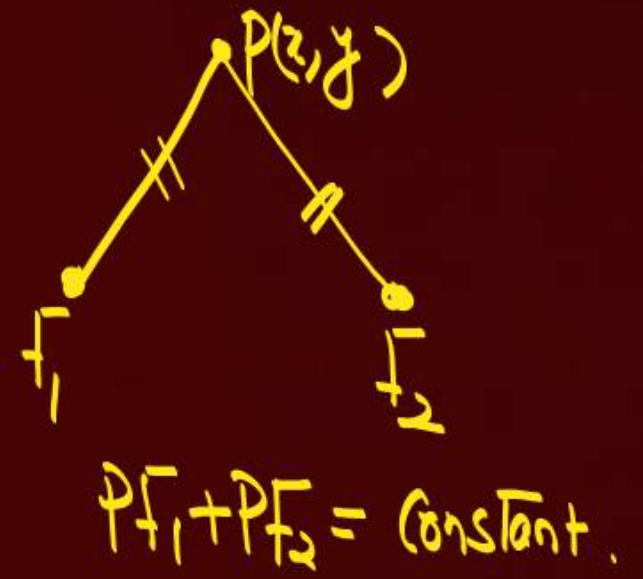
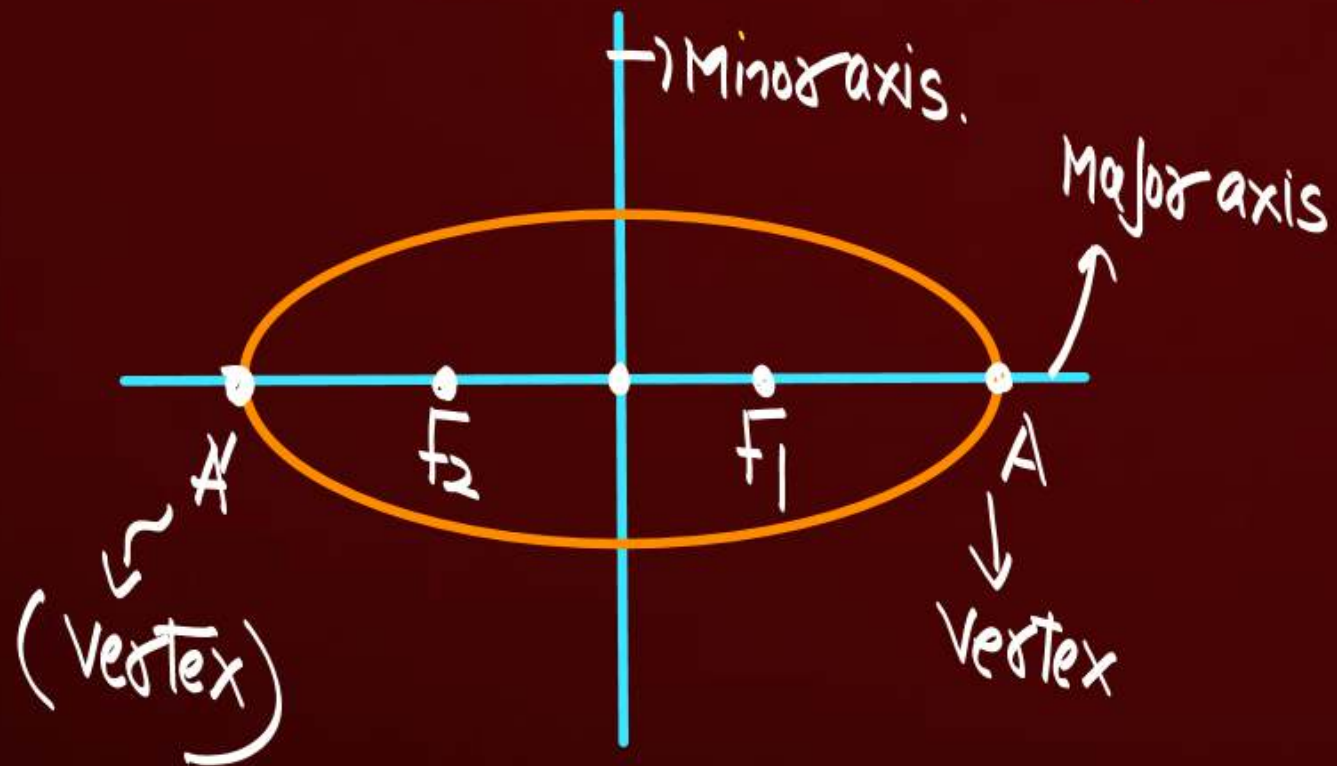
Main Facts about the parabola

Forms of Parabolas	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
Axis	$y = 0$	$y = 0$	$x = 0$	$x = 0$
Directrix	$x = -a$	$x = a$	$y = -a$	$y = a$
Vertex	$(0, 0)$	$(0, 0)$	$(0, 0)$	$(0, 0)$
Focus	$(a, 0)$	$(-a, 0)$	$(0, a)$	$(0, -a)$
Length of latus rectum	$4a$	$4a$	$4a$	$4a$
Equations of latus rectum	$x = a$	$x = -a$	$y = a$	$y = -a$

Ellipse

The collection of points in a plane such that the sum of its distances from two fixed points is constant is known as an ellipse.

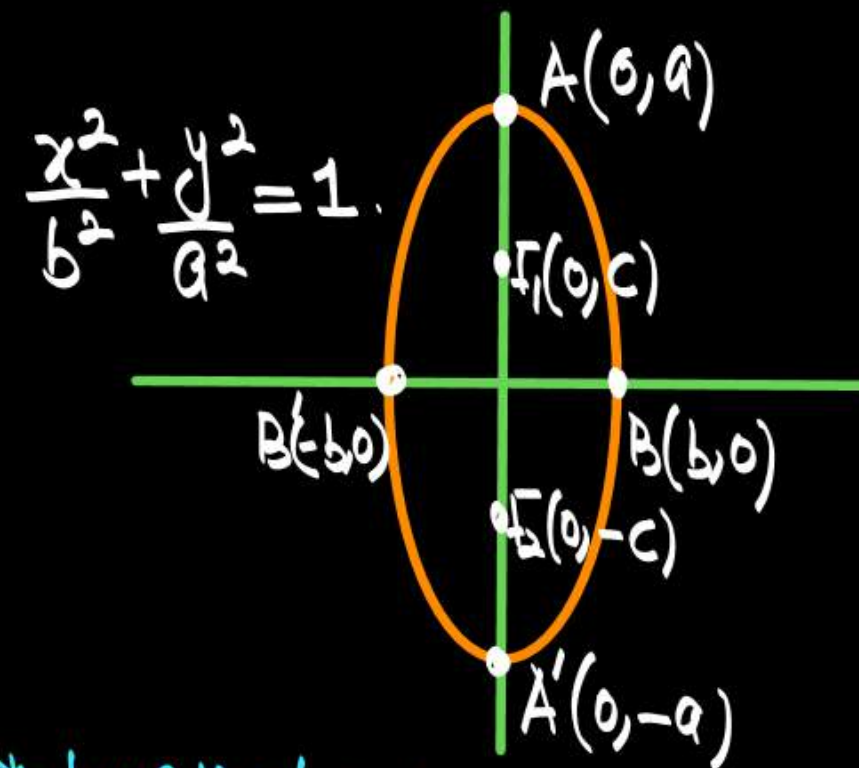
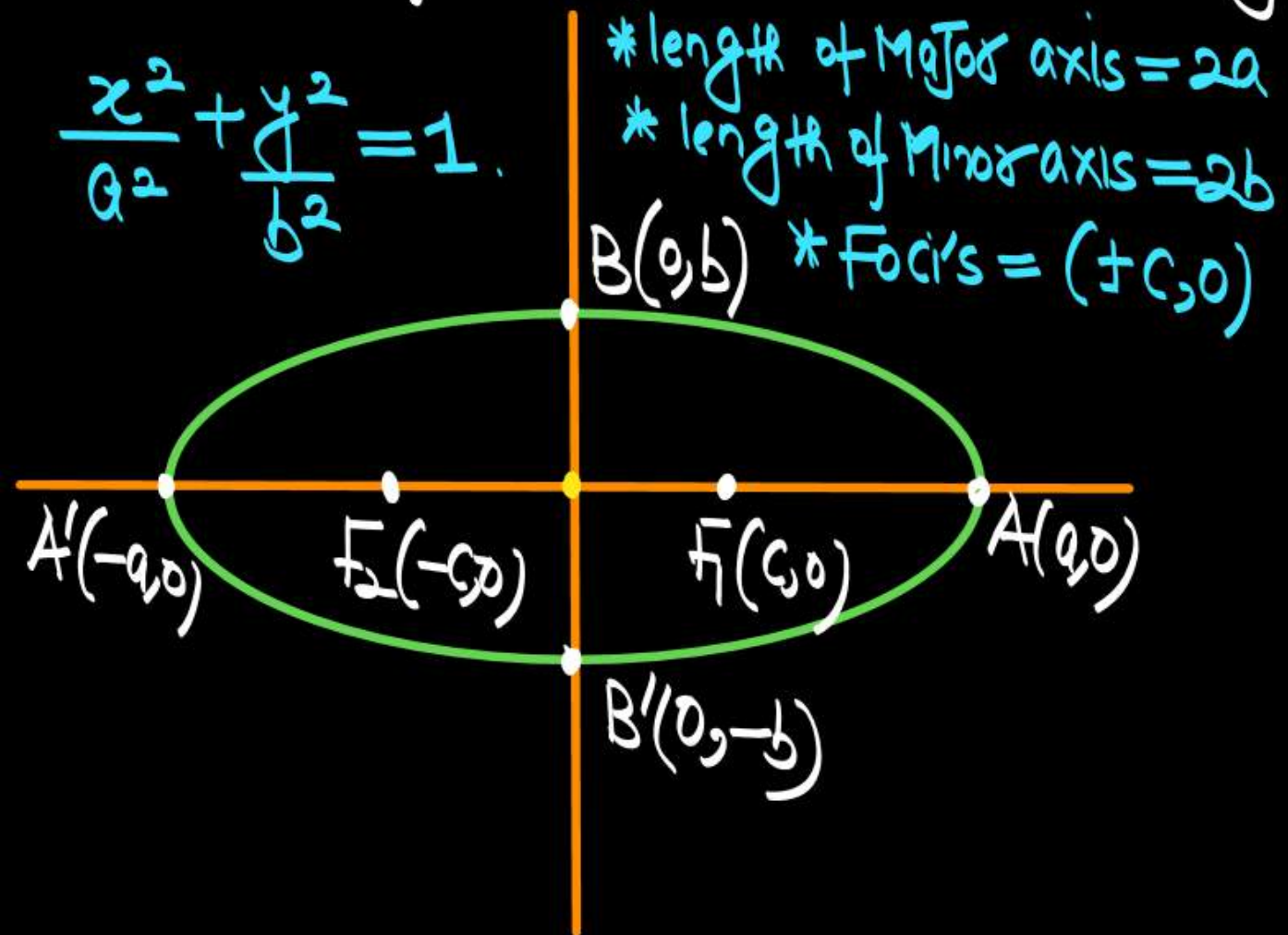
* Fixed points are known as the foci of the ellipse.



Equation of ellipse in the standard form

* Major axis and minor axis are taken along the Co-ordinate axes.

* Centre of the ellipse is taken at the origin.



* length of Major axis = $2a$
 * length of Minor axis = $2b$
 * Foci's = $(0, \pm c)$

* $c = \sqrt{a^2 - b^2}$

* length of latus Rectum = $\frac{2b^2}{a}$

* eccentricity,
 $e = \frac{c}{a}$ ($e > 0$)
 * $e < 1$

EXAMPLE



Find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the latus rectum of the ellipse.

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

(Sol) Given ellipse is:

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$\text{Here } a^2 = 25, b^2 = 9 \\ \Rightarrow a = 5, b = 3.$$

$$\text{So } c = \sqrt{a^2 - b^2} = \sqrt{25 - 9} \\ \Rightarrow c = 4 \text{ units.}$$

$$\text{Therefore foci} = (\pm c, 0) = (\pm 4, 0)$$

$$\text{Vertices} = (\pm a, 0) = (\pm 5, 0)$$

$$\text{length of major axis} = 2a = 10.$$

$$\text{length of minor axis} = 2b = 6$$

$$\text{eccentricity, } e = \frac{c}{a} = \frac{4}{5}.$$

$$\begin{aligned} \text{length of latus Rectum} &= \frac{2b^2}{a} \\ &= \frac{2 \times 9}{5} \\ &= \frac{18}{5} \text{ units.} \end{aligned}$$

EXAMPLE



Find the coordinates of the foci, the vertices, the lengths of major and minor axes and the eccentricity of the ellipse $9x^2 + 4y^2 = 36$.

$$(0, \pm\sqrt{5})$$

$$c = \sqrt{9 - 4} = \sqrt{5}$$

$$\frac{9x^2}{36} + \frac{4y^2}{36} = \frac{36}{36}$$

$$\Rightarrow \frac{x^2}{4} + \frac{y^2}{9} = 1$$

$$a = 3$$

$$b = 2$$

EXAMPLE



Find the equation of the ellipse, with major axis along the x -axis and passing through the points $(4, 3)$ and $(-1, 4)$.

H/W

Main Facts of Ellipse

Forms of the ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$
	$a > b$	$a > b$
Equation of major axis	$y = 0$	$x = 0$
Length of major axis	$2a$	$2a$
Equation of Minor axis	$x = 0$	$y = 0$
Length of Minor axis	$2b$	$2b$
Directrices	$x = \pm \frac{a}{e}$	$y = \pm \frac{a}{e}$
Equation of latus rectum	$x = \pm ae$	$y = \pm ae$
Length of latus rectum	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$
Centre	$(0, 0)$	$(0, 0)$

QUESTION



In each of the Exercises 1 to 9, find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

#

$$\frac{x^2}{36} + \frac{y^2}{16} = 1$$

QUESTION



In each of the Exercises 1 to 9, find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

h/w

$$\frac{x^2}{25} + \frac{y^2}{100} = 1$$

QUESTION



In each of the Exercises 1 to 9, find the coordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse.

H/w

$$36x^2 + 4y^2 = 144$$

QUESTION



In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

Vertices $(\pm 5, 0)$, foci $(\pm 4, 0)$

(Sol) Here $a = 5$ & $c = 4$.

$$\Rightarrow c^2 = 16$$

$$\Rightarrow a^2 - b^2 = 16$$

$$\Rightarrow b^2 = a^2 - 16 = 25 - 16 = 9$$

So Req. =

$$\frac{x^2}{25} + \frac{y^2}{9} = 1$$

QUESTION



In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

Length of major axis 26, foci $(\pm 5, 0)$

$$2a = 26$$

$$a = 13$$

$$c = 5$$

$$a^2 - b^2 = c^2$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

QUESTION



In each of the following Exercises 10 to 20, find the equation for the ellipse that satisfies the given conditions:

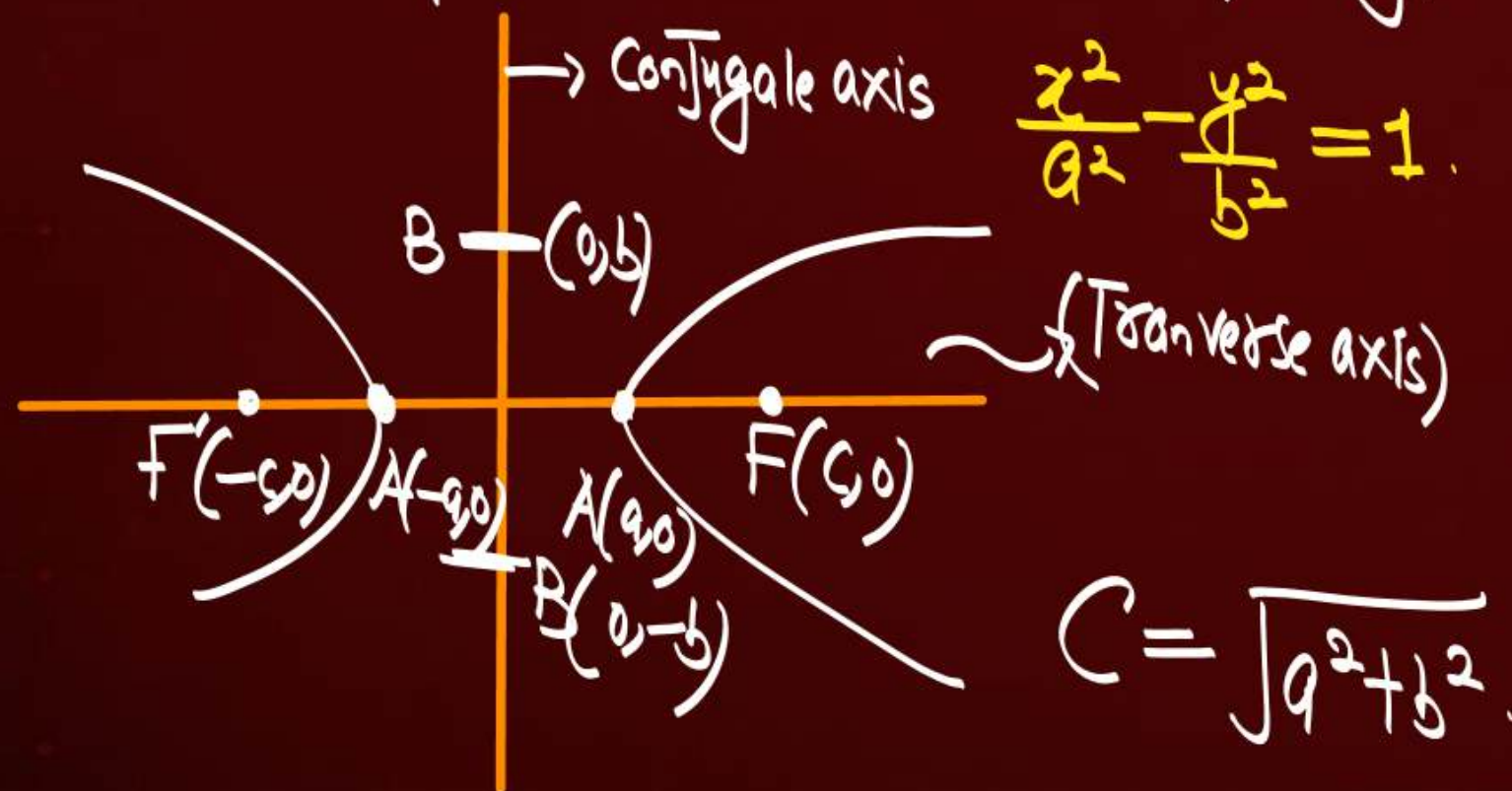
Centre at $(0, 0)$, major axis on the y -axis and passes through the points $(3, 2)$ and $(1, 6)$.

h/w

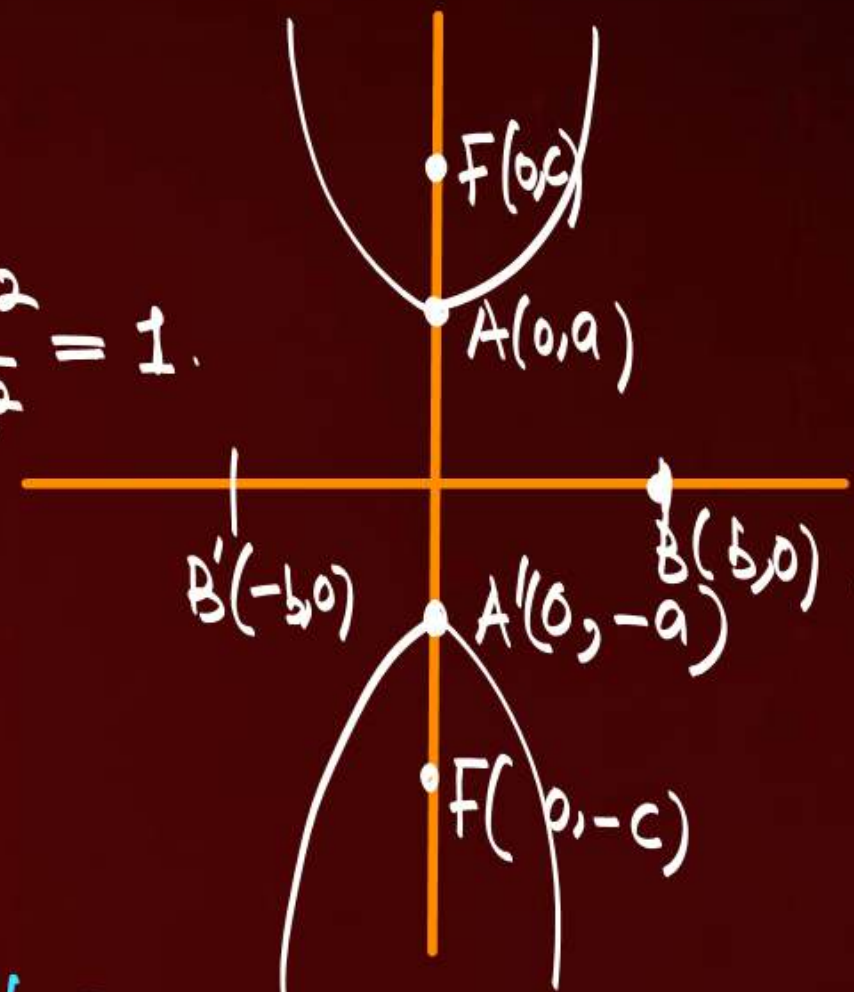
Hyperbola

The Collection of points in a Plane Such that the difference of it's distances from two fixed points is Constant is known as Hyperbola.

* The fixed points are known as foci of the hyperbola.



$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$



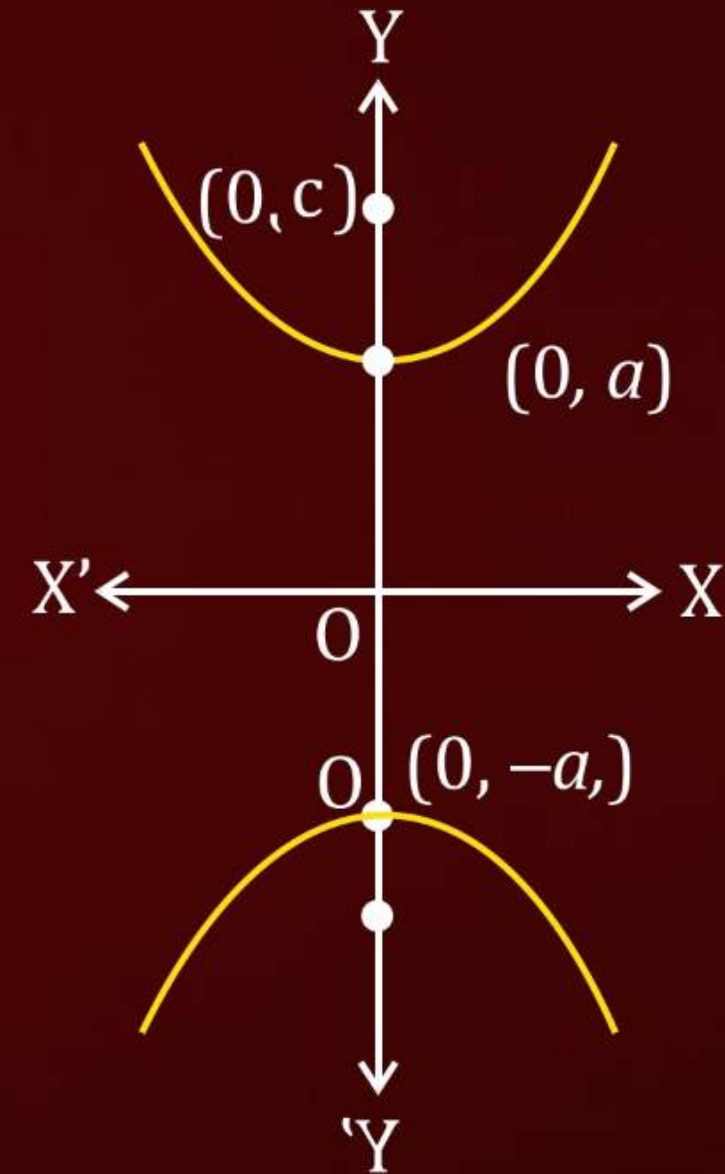
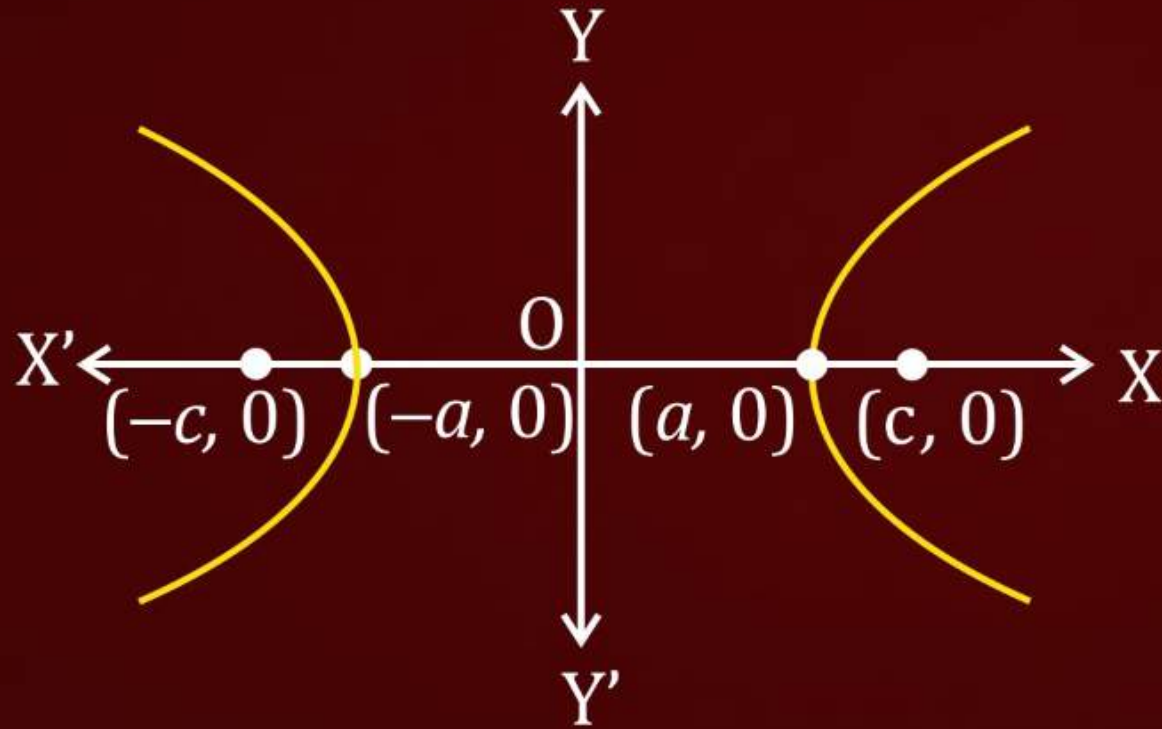
* length of Transverse axis = $2a$

* length of Conjugate axis = $2b$

* length of latus rectum = $\frac{2b^2}{a}$

* eccentricity, $e = \frac{c}{a}$
 $e > 1$

Standard equation of Hyperbola



EXAMPLE



Find the coordinates of the foci and the vertices, the eccentricity, the length of the latus rectum of the hyperbolas:

(i) $\frac{x^2}{9} - \frac{y^2}{16} = 1$

(ii) $y^2 - 16x^2 = 16$

(Sol) (i) Given Hyperbola is:

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$\Rightarrow \frac{x^2}{3^2} - \frac{y^2}{4^2} = 1$$

So, $a = 3$ & $b = 4$.

Here $c = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = 5$.

$$\Rightarrow \text{foci} = (\pm 5, 0)$$

$$\text{Vertices} = (\pm a, 0) = (\pm 3, 0)$$

$$\text{eccentricity, } e = \frac{c}{a} = \frac{5}{3}$$

$$\text{length of latus rectum} = \frac{2b^2}{a} = \frac{32}{3}$$

EXAMPLE

(Important).

Find the equation of the hyperbola where foci are $(0, \pm 12)$ and the length of the latus rectum is 36.

(Sol) Given that $c = 12$ and $\frac{2b^2}{a} = 36$.

$$\Rightarrow \sqrt{a^2 + b^2} = 12 \text{ \& } 2b^2 = 36a$$

$$\Rightarrow \sqrt{a^2 + b^2} = 12 \text{ \& } b^2 = \frac{36a}{2}$$

$$\Rightarrow a^2 + b^2 = 144 \text{ --- (1)}$$

$$\text{ \& } b^2 = 18a \text{ --- (2)}$$

Using (2) in (1), we get;

$$a^2 + 18a = 144$$

$$\Rightarrow a^2 + 18a - 144 = 0.$$

$$\Rightarrow a^2 + 24a - 6a - 144 = 0.$$

$$\Rightarrow a(a + 24) - 6(a + 24) = 0.$$

$$\Rightarrow (a - 6)(a + 24) = 0.$$

$$\Rightarrow a = 6 \Rightarrow a^2 = 36.$$

$$\text{So, } b^2 = 18 \times 6$$

$$\Rightarrow b^2 = 108.$$

Therefore the ^{req.} equation of the hyperbola is:

$$\frac{y^2}{36} - \frac{x^2}{108} = 1.$$

$$\Rightarrow y^2 - \frac{x^2}{3} = 36$$

$$\Rightarrow \boxed{3y^2 - x^2 = 108}.$$

Hence the result.

Main Facts about the Hyperbola

Forms of the Hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$
Equation of transverse axis	$y = 0$	$x = 0$
Equation of conjugate axis	$x = 0$	$y = 0$
Length of transverse axis	$2a$	$2a$
Foci	$(\pm ae, 0)$	$(0, \pm ae)$
Equation of latus rectum	$x = \pm ae$	$y = \pm ae$
Length of latus rectum	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$
Centre	$(0, 0)$	$(0, 0)$

QUESTION



In each of the Exercises 1 to 6, find the coordinates of the foci and the vertices, the eccentricity and the length of the latus rectum of the hyperbolas.

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

H/W

QUESTION



In each of the Exercises 1 to 6, find the coordinates of the foci and the vertices, the eccentricity and the length of the latus rectum of the hyperbolas.

$$\frac{x^2}{9} - \frac{y^2}{27} = 1$$

H/W

QUESTION



In each of the Exercises 1 to 6, find the coordinates of the foci and the vertices, the eccentricity and the length of the latus rectum of the hyperbolas.

$$9y^2 - 4x^2 = 36$$

$$\frac{9y^2}{36} - \frac{4x^2}{36} = 1$$

$$\frac{y^2}{4} - \frac{x^2}{9} = 1$$

$$(0, \pm c)$$

$$(0, \pm a)$$

$$\frac{2b^2}{a}$$

$$c = \sqrt{a^2 + b^2}$$

QUESTION



In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1.$$

Vertices $(0, \pm 5)$, foci $(0, \pm 8)$

$$a = 5, c = 8$$

$$c^2 = 64$$

$$a^2 + b^2$$

$$25 + b^2 = 64$$

$$b^2 = 64 - 25.$$

QUESTION

In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

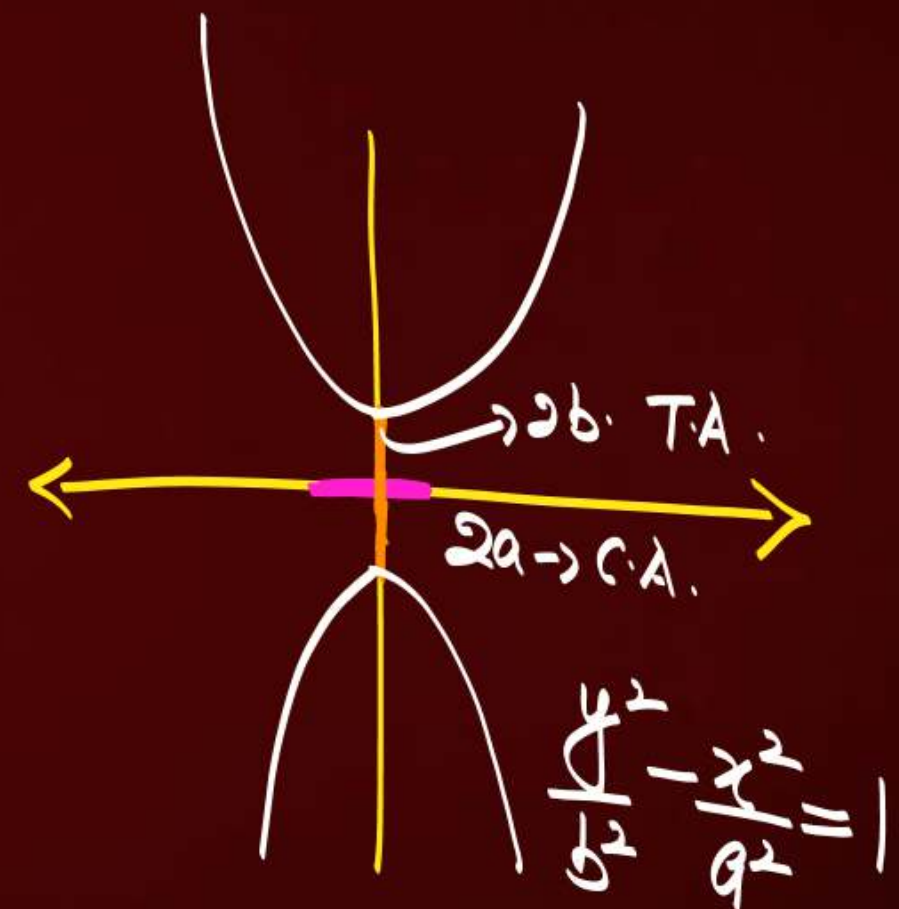
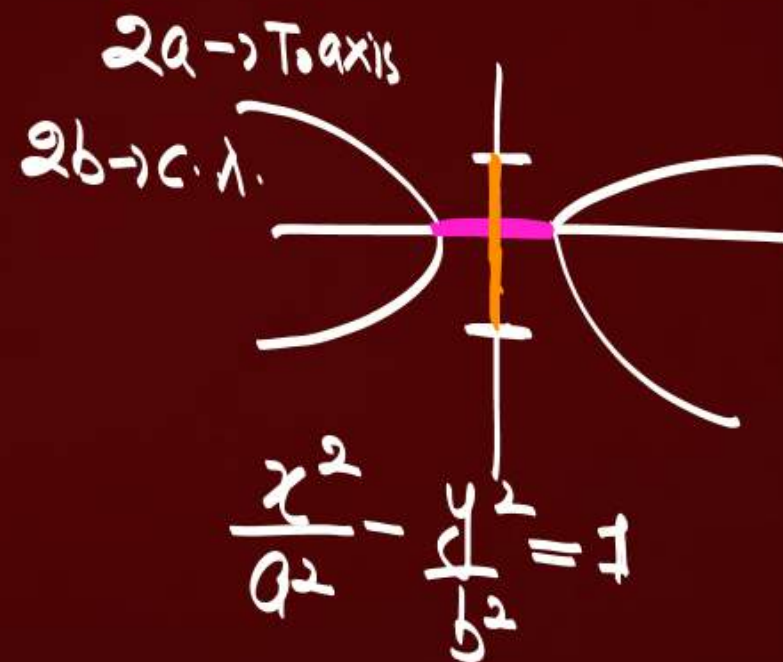
Foci $(0, \pm 13)$, the conjugate axis is of length 24.

$$c = 13$$

$$a^2 + b^2 = 169$$

$$2b = 24$$

$$b = 12$$



QUESTION



In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

Foci $(\pm 4, 0)$, the latus rectum is of length 12.

H/W

QUESTION



In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Vertices $(\pm 4, 0)$, $e = \frac{4}{3}$.

$$a = 4, \quad \frac{c}{a} = \frac{4}{3},$$

$$c = \frac{16}{3}.$$

QUESTION



In each of the Exercises 7 to 15, find the equations of the hyperbola satisfying the given conditions.

Foci $(0, \pm \sqrt{10})$, passing through $(2, 3)$

H/W (Important)

Thank You

